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Reactive oxygen species generation induced by coupling *Phanerochaete chrysosporium* with intermittent micro-aeration to enhance methane production

Xuepeng Wang, Zhenru Wang, Ying Deng, Haohao Mao, Qilin Yu, Yaobin Zhang *

Dalian University of Technology School of Environmental Science and Technology No.2 Linggong Road, Ganjingzi District, Dalian, Liaoning 116024, China

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ABSTRACT

Anaerobic digestion is constrained by the rate-limiting step of hydrolysis and acidogenesis of macromolecular organic matter, which significantly affects the overall process efficiency and system stability. This study introduced aerobic fungi *Phanerochaete chrysosporium* into an anaerobic digestion system combined with intermittent micro-aeration to induce the generation of extracellular reactive oxygen species, thereby accelerating the decomposition of organic solid waste and promoting methane production. The results showed that under an optimal aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day, the intermittent micro-aeration (IMA) group enhanced cumulative methane production by 17.22 %, while the combined application of IMA with *P. chrysosporium* (IMA-P) group increased by 26.36 %, both relative to the non-aerated control group lacking *P. chrysosporium* supplementation. The implementation of intermittent micro-aeration induced periodic oscillations in oxidation-reduction potential, ranging from −460 mV to −300 mV, which subsequently triggered the generation of •OH at concentrations between 35.84 μM and 65.29 μM. Under the limited oxygen condition, *P. chrysosporium* secreted lignocellulolytic enzymes and enhanced •OH generation to accelerate the degradation of lignocellulose. Microbial community analysis revealed that *P. chrysosporium* could proliferate in the anaerobic digester under such intermittent micro-aeration conditions. The enrichment of methanogens capable of coping with O_2 stress, along with the elevated abundance of antioxidant enzyme genes, might account for the effective continuation of methanogenesis under intermittent micro-aeration conditions. This study offered a feasible method to improve the anaerobic digestion of complex substrates through the combination of *P. chrysosporium* supplementation and intermittent micro-aeration.

1. Introduction

Anaerobic digestion has been extensively employed as a sustainable biotechnology for resource recovery and energy generation from organic waste materials. However, the anaerobic digestion of solid organic wastes is constrained by the rate-limiting hydrolysis and acidogenesis stages, particularly in the degradation of recalcitrant lignocellulosic components [1]. The structural complexity and chemical stability of lignocellulose result in low biogas production efficiency, thereby significantly affecting the overall process efficiency and system stability [2].

White-rot fungi are specialized aerobic microorganisms that exhibit remarkable lignocellulose-degrading capacities through the production of extracellular oxidative enzymes [3,4]. Under aerobic conditions, they secrete ligninolytic enzymes such as laccase (Lac), manganese peroxidase (MnP), and lignin peroxidase (LiP), which oxidize a wide range of substrates, including lignin, phenols, and other aromatic compounds [3,5]. Furthermore, the ligninolytic system of white-rot fungi generates reactive oxygen species (ROS), such as extracellular hydroxyl radicals (•OH), which effectively promote lignin degradation when ligninolytic enzymes cannot penetrate plant cell walls [6]. Specifically, the lignocellulose-oxidizing enzymes secreted by white-rot fungi catalyze the reduction of oxygen (O_2) to superoxide radicals ($•O_2^-$), which undergo dismutation to form hydrogen peroxide (H_2O_2) ($•O_2^- + HO_2• + H^+ \rightarrow O_2 + H_2O_2$). Subsequently, H_2O_2 reacts with Fe(II) via Fenton-like reactions ($Fe(II) + H_2O_2 \rightarrow Fe(III) + •OH + OH^-$) to generate •OH [7,8]. As a potent oxidant with a high redox potential, •OH (an oxidation potential of 2.8 V vs. Standard Hydrogen Electrode) can disrupt the

* Corresponding author.
E-mail address: zhangyb@dlut.edu.cn (Y. Zhang).

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complex structure of lignocellulose [9]. Extensive research has demonstrated that white-rot fungi significantly enhance lignocellulose biodegradation and organic pollutant removal in aerobic systems [7,10]. These unique capabilities have led to their successful application in various biological processes, including aerobic composting processes for organic waste treatment and pre-treatment systems to improve the biodegradability of lignocellulosic materials prior to anaerobic digestion [4].

Traditionally, anaerobic digestion systems have been designed to exclude O_2 due to its inhibitory effects on obligate anaerobic microorganisms such as methanogens. However, recent studies have demonstrated that a controlled O_2 supply can enhance the anaerobic digestion performance of organic waste. For example, Fu et al. reported that intermittent aeration improved the cumulative methane yield and volatile suspended solids (VS) removal rate of corn straw by 10.3 % and 16.5 %, respectively [11]. In practice, air mixing replacing mechanical mixing has been a common strategy for enhancing organic solid waste treatment in biogas engineering [12,13]. Under optimal conditions, micro-aeration does not inhibit methanogens but improves process efficiency [11,14]. The enhanced performance is generally attributed to increased microbial diversity in anaerobic systems, accelerating the hydrolysis of complex organic matter [15,16].

Recent studies have revealed that introducing intermittent micro-aeration into anaerobic digestion systems can induce the generation of microbial extracellular ROS [17]. ROS are widely present in anoxic-oxic conditions, such as paddy soils, intertidal zones, and aerobic composting systems [18–20]. Specifically, under anaerobic conditions, microbial extracellular respiration transfers electrons to external electron acceptors (e.g., Fe(III)), generating Fe(II). In aerobic environments, O_2 accepts electrons from microorganisms or Fe(II), producing ROS such as $\bullet O_2^-$ and H_2O_2, which then react with Fe(II) via Fenton-like reactions to generate •OH. These ROS, mediated by microorganisms under periodic anoxic-oxic conditions, play a critical role in substance transformation and cycling in natural ecosystems and engineered biosystems [17,21,22]. For instance, intermittent micro-aeration supply in anaerobic systems induced •OH generation, and enhanced the degradation of pharmaceuticals such as carbamazepine and sulfamethoxazole [17].

Considering that white-rot fungi can generate extracellular ROS, a result of extracellular O_2 accepting the intracellular electron, it is reasonable to hypothesize that other extracellular electron acceptors such as electrodes or Fe oxides (ferrihydrite) with suitable oxidation-reduction potential (ORP) may substitute the extracellular O_2 to maintain the metabolism of white-rot fungi under anaerobic conditions. The redox fluctuations induced by intermittent micro-aeration in anaerobic systems may allow white-rot fungi to proliferate and play a role in the anaerobic digestion of complex organic waste. However, rare research has been reported on white-rot fungi-enhanced micro-aerobic anaerobic digestion except for a recent study by Gong et al., who found that *Phanerochaete chrysosporium* added to a micro-aerobic anaerobic digester could secrete Lac, MnP, and LiP to accelerate straw liquefaction and promote methane production [2]. However, whether white-rot fungi induce the generation of ROS in anaerobic digestion systems is unknown. It is critical to elucidate the generation of ROS by white-rot fungi in oxygen-limited environments, which deepens the understanding of their role in complex organic matter degradation under natural and engineered biosystems.

This study aimed to accelerate lignocellulose degradation and enhance methane production by introducing white-rot fungi coupled with intermittent micro-aeration to induce the generation of ROS. *P. chrysosporium* was used here because it is commonly considered a model fungus for white rot fungi and is an efficient delignifier [23,24]. Particularly, this study was conducted to: (1) investigate the effects of white-rot fungi coupled with intermittent micro-aeration on anaerobic digestion; (2) analyze redox oscillations and ROS generation; and (3) determine the optimal aeration intensity for white-rot fungi proliferation through microbial community analysis.

2. Materials and methods

2.1. Experimental materials

The inoculum sludge and cow manure were collected from an organic waste treatment company in Hangzhou. After being collected from a full-scale plant, the inoculum sludge was cultivated with cow manure as the substrate in a lab-scale semi-continuous reactor to maintain high methanogenic activity. The raw cow manure was stored at 4 °C before use to maintain its stable properties. The total suspended solids (TS) of the inoculum sludge were 19.35 g/L, and the VS was 6.03 g/L. The TS of the cow manure was 35.51 g/L, and the VS was 26.80 g/L. *P. chrysosporium* was purchased from the Beina Biological Engineering and Technology Research Center and inoculated in potato dextrose agar medium at 30 °C. After culturing the spore suspensions of *P. chrysosporium* ($OD_{600} = 1$) for two week, the mycelium was harvested, dewatered via gauze filtration, and stored for subsequent experiments.

2.2. Batch experiment

The experiment was conducted in batch mode using 500 mL threaded bottles. Three groups were conducted, i.e., the intermittent micro-aeration group (referred to as IMA group), the intermittent micro-aeration with *P. chrysosporium* addition group (referred to as IMA-P group), and the control group with no IMA and *P. chrysosporium*. Each group has three parallels. The experiment was run in three stages, each stage lasted for 36 days. At the beginning of each stage, 100 mL of inoculum (or digested liquid from the previous stage) was added, along with 350 mL of fresh cow manure (6 % solids content), and ferrihydrite (10 mM Fe(III)) was introduced as an iron source. Intermittent micro-aeration was performed daily by injecting a certain air volume into the reactors using a syringe, i.e., Stage 1 and 2: 20 mL/day (4 %($V_{Aeration}$:$V_{Reactor}$)/day), Stage 3: 40 mL/day (8 %($V_{Aeration}$:$V_{Reactor}$)/day). The selection of the intensity of intermittent micro-aeration is detailed in Text S1. The IMA-P group only received an addition of *P. chrysosporium* mycelium (2 g wet weight/450 mL) at the start of the first stage. To avoid the influence of abiotic factors, the control and IMA groups were supplemented with an equivalent amount of inactivated *P. chrysosporium* mycelium. The prepared anaerobic bottles were placed in a constant-temperature incubator at 37 °C. During the experiment, sludge samples were collected using a syringe from rubber tubes equipped with valves, and the produced biogas was collected using gas bags.

2.3. ROS measurement

The •OH generation was determined by an electron paramagnetic resonance (EPR) spectrometer and quantified by coumarin, the H_2O_2 was determined by cerium sulfate spectrophotometry as described in Text S2 [25,26].

2.4. Characterization and analysis

TS, VS, and biogas content and volume were analyzed as previously reported and detailed in Text S3. The capacitance and resistance of the sludge were determined according to the previous study and detailed in Text S4 [27].

Cellulose, hemicellulose, and lignin were determined using the Alva semi-automatic crude fiber analyzer F1600 (Text S5). ORP and oxygen content in the headspace were evaluated by an ORP composite electrode (STARTER, 3100, USA) and a fiber-optic oxygen transmitter (FiBox3, PreSens, Regensburg, Germany), respectively. The activity of enzymes (Lac, LiP, and MnP) was quantified by enzyme-linked immunosorbent assay kits (Text S6).

The morphology of sludge after digestion was observed using a scanning electron microscope (SEM, HITACHI SU5000, China).

Functional groups of the sludge samples were evaluated by a Fourier-transform infrared (FTIR) spectrometer (ThermoFisher, 6700, Germany) with a scanning range from 4000 to 400 cm^{-1}. The composition of dissolved organic matter in the suspension was determined using a three-dimensional fluorescence spectrometer (Agilent, Cary Eclipse, Malaysia).

2.5. Microbial community analysis

The fungal community of sludge was analyzed via high-throughput sequencing by Sangon Biotech (Shanghai) Co., Ltd., China. The metagenomic analysis was conducted by Shanghai GeneCowin Biotechnology Co., Ltd., China. The detailed methods are provided in Text S7.

3. Results and discussion

3.1. Performance of anaerobic digestion

Batch experiments with the control group, the intermittent micro-aeration (IMA) group and the intermittent micro-aeration with *P. chrysosporium* addition (IMA-P) group were conducted over three stages (3 × 36 days) under different aeration intensities. Ferrihydrite (10 mM Fe(III)) was added as an iron source at the beginning of each stage. The results demonstrated that *P. chrysosporium* coupled with intermittent micro-aeration effectively enhanced methane production from anaerobic digestion of organic solid waste. During stage 1–2 (0–72 days), with an aeration intensity of 4 % ($V_{Aeration}$:$V_{Reactor}$)/day, the cumulative methane production in the IMA group ranged from 126.35 to 134.72 mL/gVS, 5.91 %–7.60 % higher than the control group (119.30–125.20 mL/gVS). In contrast, the IMA-P group exhibited higher cumulative methane production (126.61–141.50 mL/gVS), achieving a 6.13 %–13.02 % enhancement over the control group (Fig. 1A). These results indicated that intermittent micro-aeration combined with *P. chrysosporium* addition promoted methane production.

Given the aerobic characteristics of *P. chrysosporium*, the limited oxygen supply might limit its activity. During stage 3 (72–108 days), the aeration intensity was adjusted to 8 % ($V_{Aeration}$:$V_{Reactor}$)/day. The IMA and IMA-P groups achieved cumulative methane production of 137.76 mL/gVS and 148.57 mL/gVS, respectively, corresponding to a 17.22 % and 26.36 % increase over the control group (118.73 mL/gVS). Under similar operating conditions (25–39 °C), the specific methane production obtained in this study was consistent with the literature range (49.54–409.76 mL/g VS) (Table S1). Moreover, the specific methane production in this work exceeded most of the previously reported values. Additionally, the methane production of the control group (118.73 mL/g VS) in stage 3 exhibited a slight decrease compared to the preceding two stages (119.30 mL/g VS, 125.20 mL/g VS, respectively), which might be attributed to the loss of functional microorganisms during substrate replacement in batch experiments. On the contrary, both the IMA and IMA-P groups displayed a consistent upward trend in methane production throughout stages 1 to 3. This result also indicated that both the IMA group and the IMA-P group demonstrated higher anaerobic digestion efficiency and greater stability in long-term operation.

Intermittent micro-aeration combined with *P. chrysosporium* addition improved organic substrate degradation. Sludge sampled on day 108 was analyzed for waste reduction. As shown in Fig. 1B, the TS removal rate of the IMA group (43.06 %) and the IMA-P group (41.44 %) surpassed the control group (33.85 %) by 9.21 and 7.60 percentage points, respectively. Similarly, VS removal was enhanced in both micro-aeration groups. Additionally, cows digest the readily degradable portions of feed (grass or silage) through rumen microbes, leaving cow manure with a fibrous nature. Specifically, the raw cow manure contained 26.28 % ± 0.50 % hemicellulose, 29.41 % ± 0.67 % cellulose, and 14.07 % ± 1.14 % lignin in this study, respectively, consistent with the contents reported in the literature (Table S2). The complex structures of recalcitrant components (cellulose and lignin) in cow manure impede the hydrolysis process during anaerobic digestion [28,29]. After the treatment, the IMA-P group exhibited the lowest lignocellulose content at 60.71 % (hemicellulose: 26.27 %, cellulose: 23.83 %, lignin:

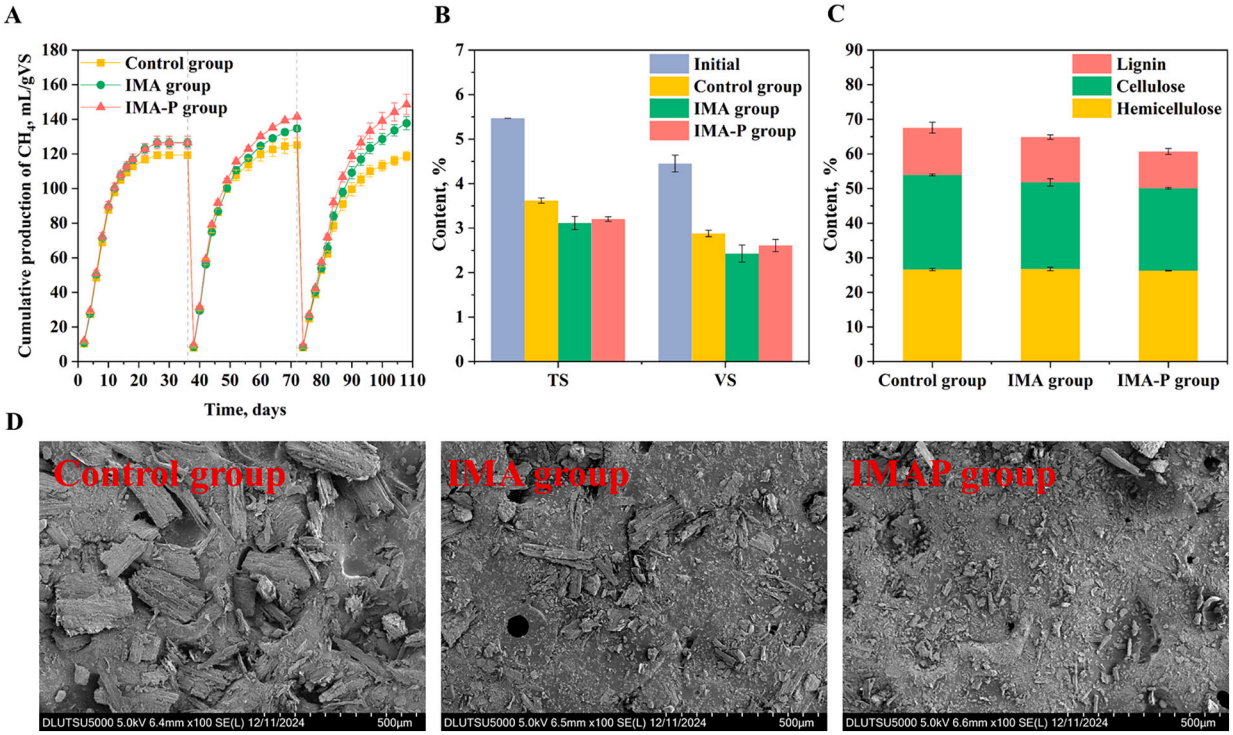


Fig. 1. The performance of anaerobic digestion. (A) Cumulative methane production. (B) Changes of TS and VS before and after digestion. (C) Lignocellulose content after treatment. (D) The SEM images of the sludge after treatment.

10.61 %), followed by the IMA group with 64.85 % (hemicellulose: 26.76 %, cellulose: 25.04 %, lignin: 13.06 %), while the control group showing the highest residual lignocellulose at 67.58 % (hemicellulose: 26.58 %, cellulose: 27.37 %, lignin: 13.63 %) (Fig. 1C). These results suggested that intermittent micro-aeration combined with *P.chrysosporium* accelerated the decomposition of lignocellulose [2]. Accordingly, SEM showed that undigested straw-like residues were abundant in the sludge of the control group, while smaller particle sizes were observed in the IMA and IMA-P groups (Fig. 1D).

3.2. Redox oscillations and ROS generation

At an aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day, ORP and headspace O_2 contents were monitored. Results revealed that ORP rapidly increased to approximately −300 mV within 1 h after aeration, followed by a gradual decline to pre-aeration levels (about −460 mV). Concurrently, headspace O_2 contents initially rose and declined after aeration (Fig. 2), indicating oxygen diffusion into the liquid phase and subsequent microbial consumption. The ORP fluctuations confirmed that intermittent micro-aeration induced redox oscillations in anaerobic digestion, creating favorable conditions for ROS generation [17].

Notably, *P. chrysosporium* accelerated oxygen utilization after aeration. For example, ORP in the IMA group recovered to −429.5 mV (n = 4) 6 h after aeration, while the IMA-P group achieved a faster recovery to −461.25 mV (n = 4). This was corroborated by headspace O_2 dynamics: O_2 contents in the IMA-P group (0.542 % and 0.036 % at 3 h and 6 h, respectively) were significantly lower than those in the IMA group (0.912 % and 0.070 % at 3 h and 6 h, respectively) after aeration.

Redox oscillations typically induce ROS production [18,30]. To validate ROS generation triggered by intermittent micro-aeration, ROS quantification was conducted at peak oxidation-reduction potential (1 h post-aeration) within the bulk liquid phase. H_2O_2 was quantified using cerium sulfate spectrophotometry, and the results showed an average concentration of 24.19 ± 6.75 μM in the IMA-P group, 90.98 % higher than the IMA group (12.67 ± 3.78 μM), while the control group showed negligible detection (Fig. 3A).

Using DMPO (5,5-Dimethyl-1-Pyrroline-N-Oxide) as a spin-trapping agent to identify •OH, characteristic DMPO/•OH adduct signals (hyperfine coupling constants: α_N = 14.89 G, α_H^{β} = 14.89 G) were detected in both intermittent micro-aeration groups but not in the control (Fig. 3B). The DMPO/•OH signal intensity in the IMA-P group was significantly higher than that in the IMA group, indicating greater •OH production [31,32]. For quantitative analysis, 100 μM coumarin was added to the reactors to trap •OH and form 7-hydroxylcoumarin [33]. The results showed cumulative •OH levels within 1 h after micro-aeration in the IMA-P group was 65.29 ± 6.93 μM, 82.18 % higher

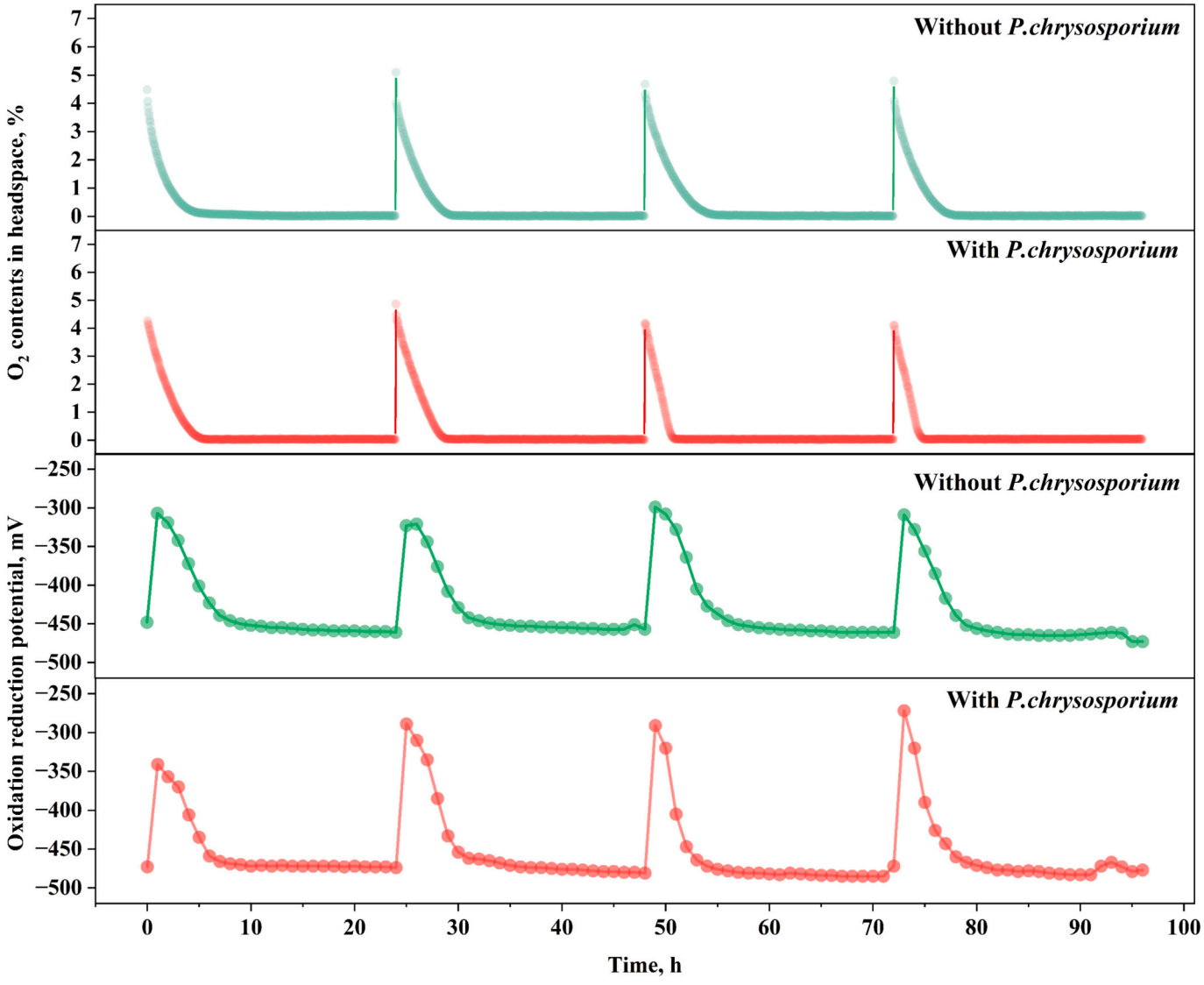


Fig. 2. Redox oscillations induced by intermittent micro-aeration with an aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day.

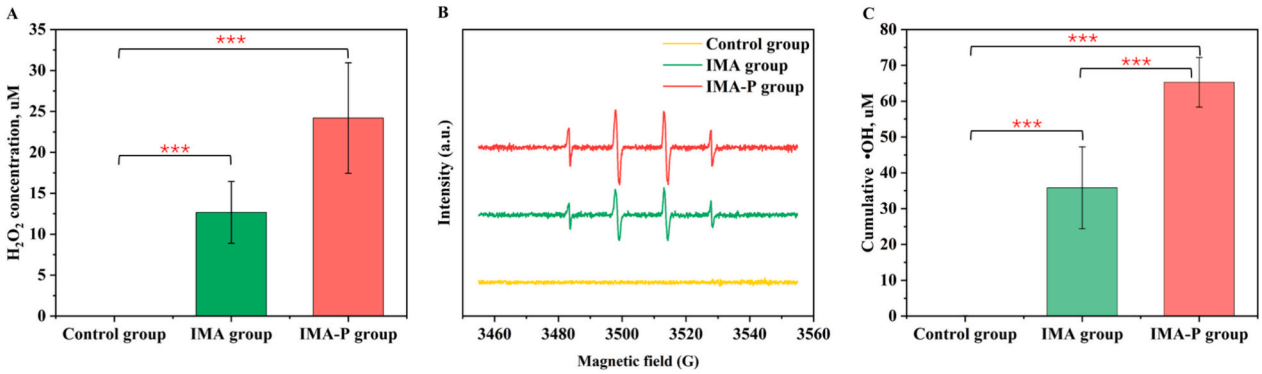


Fig. 3. Reactive oxygen species generation after intermittent micro-aeration. (A) Concentration of H_2O_2. (B) EPR spectra of •OH after intermittent micro-aeration. (C) Cumulative •OH content. The red stars indicate significant differences between the two groups ($p < 0.05$).

than that in the IMA group (35.84 ± 11.43 μM), while almost no •OH was detected in the control group (Fig. 3C). The detection of H_2O_2 and •OH in anaerobic digesters indicated that redox oscillations induced Fenton-like reactions for ROS generation, furthermore, the addition of *P. chrysosporium* significantly enhanced ROS production.

As mentioned above, even without *P. chrysosporium addition*, dissolved O_2 introduced by intermittent micro-aeration could react with extracellular electrons or reduced redox mediators (e.g., Fe(II)) produced by anaerobic microbial respiration to generate ROS [17,18], through the following reaction chain:

$$O_2 + 2e^-_{(microbic)} \rightarrow \bullet O_2^- \quad (1)$$

$$Fe(II) + O_2 \rightarrow Fe(III) + \bullet O_2^- \quad (2)$$

$$Fe(II) + \bullet O_2^- + 2H^+ \rightarrow Fe(III) + H_2O_2 \quad (3)$$

$$Fe(II) + H^+ + H_2O_2 \rightarrow Fe(III) + \bullet OH + H_2O \quad (4)$$

In this study, Ferrihydrite (10 mM Fe(III)) was added as the iron source into the anaerobic digestion system at the beginning of each cycle. During organic matter oxidation, Fe(III) could be reduced to Fe (II), which subsequently reacted with H_2O_2 to generate •OH through the Fenton-like reaction. Fe(II) content in the digested sludge was determined after 108 days of operation, and ranged from 212.7 ± 3.1 mg/L to 296.9 ± 2.3 mg/L among the three groups. The presence of these Fe(II) species facilitated Fenton-like reactions to generate •OH within the intermittent micro-aeration system thereby improving the lignocellulose degradation capacity [9]. With *P. chrysosporium* addition, ROS production in the intermittent micro-aeration system was significantly enhanced. The mechanism might involve H_2O_2 generation during *P. chrysosporium* metabolism, which served as a precursor for Fenton-like reactions and further promoted •OH production [8,34].

3.3. Ligninolytic enzyme activity

White-rot fungi possess a ligninolytic enzyme system capable of secreting Lac, MnP, and LiP, which can oxidize diverse substrates, including lignin, phenols, and other aromatic compounds. The activities of Lac, MnP, and LiP were analyzed using digestate samples collected 1 h post-aeration at an aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day. Results showed negligible or minimal enzyme activity in the control group. In the IMA-P group, the activities of Lac, MnP, and LiP reached 0.18 ± 0.05 U/L, 0.78 ± 0.27 U/L, and 0.69 ± 0.14 U/L, respectively, which were 112.80 %, 424.14 %, and 23.90 % higher than the IMA group (0.09 ± 0.04 U/L, 0.15 ± 0.12 U/L, 0.58 ± 0.13 U/L) (Fig. 4). Lac, MnP, and LiP synergistically oxidize aromatic structures in lignin, cleaving C-C and C-O bonds within lignin macromolecules, thereby facilitating lignin degradation [10,35]. These results demonstrated that *P. chrysosporium* addition enhanced lignocellulolytic enzyme activity in intermittent micro-aeration systems, accelerating lignocellulose breakdown, consistent with previous studies [2].

Lignocellulolytic enzymes can catalyze the reduction of O_2 to H_2O_2, which induces Fenton-like reactions to produce •OH. The specific reaction pathway is as follows: lignin-derived quinones (e.g., benzoquinone) are reduced to hydroquinones by quinone reductase. In the presence of lignocellulolytic enzymes, hydroquinones are oxidized to free semiquinone radicals, which undergo autoxidation to regenerate quinones while producing $\bullet O_2^-$. $\bullet O_2^-$ then disproportionate to form H_2O_2, ultimately generating •OH via Fenton-like reactions (Fig. S1) [5,7]. The significantly higher activities of ligninolytic enzymes in the IMA-P group compared to the IMA group facilitated •OH production and lignocellulose degradation, in agreement with the ROS and methane production.

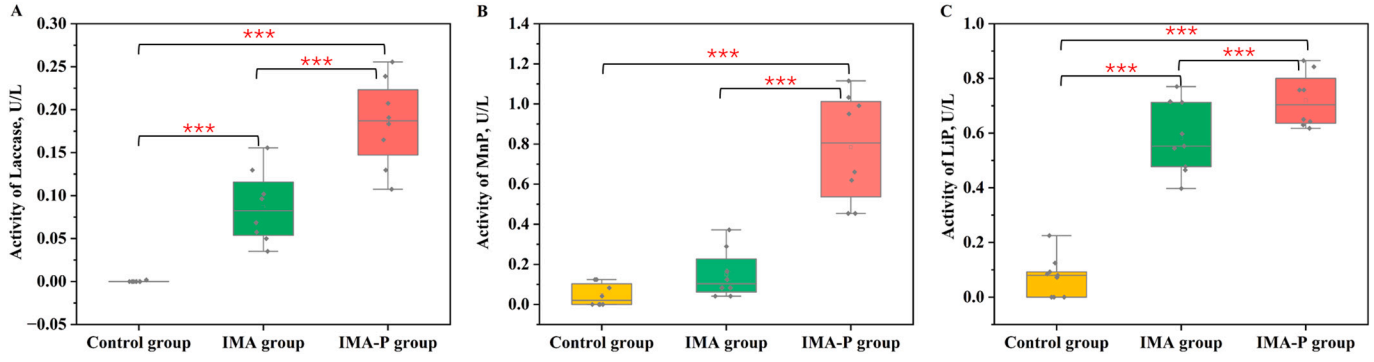


Fig. 4. The activities of ligninolytic enzymes. (A) Laccase. (B) Manganese peroxidase. (C) Lignin peroxidase. The red stars indicate significant differences between the two groups ($p < 0.05$).

3.4. Degradation of lignocellulose

Fourier-transform infrared spectroscopy (FTIR) analysis revealed structural changes in lignocellulose within digestate samples collected on day 108 (Fig. 5A). The O-H stretching vibrations (3400–3200 cm^{-1}) of lignin and cellulose showed intensified peaks in the IMA group and IMA-P group compared to the control group. It indicated that intermittent micro-aeration decreased the association of O-H to induce lignin disruption, which was likely due to the attack of •OH [36]. The band at 1600 cm^{-1} corresponding to aromatic ring stretching exhibited enhanced intensity in the IMA and IMA-P groups, suggesting destabilization of aromatic frameworks and depolymerization of lignocellulose. These observations confirmed that *P. chrysosporium* coupled with intermittent micro-aeration promoted lignocellulose degradation. Furthermore, the increased C=O peak intensity at 1650 cm^{-1} (quinone groups) in the IMA and IMA-P digestates indicated quinone formation during lignocellulose degradation [37], which might facilitate extracellular electron transfer via transformation of quinone/hydroquinones and ROS generation under the redox oscillations. Additionally, intensified C-O stretching/deformation bands (1300–800 cm^{-1}) in the IMA and IMA-P groups reflected the enhancement of cellulose/hemicellulose degradation [38].

Three-dimensional excitation-emission matrix (3D-EEM) spectroscopy further analyzed soluble organic components after the treatment. Elevated fluorescence intensities in the Ex/Em regions of 250–400/380–500 nm (fulvic acid- and humic acid-like substances) were observed in the IMA and IMA-P groups compared to the control [39]. This indicated the accelerated humification through the condensation of organic matter into macromolecular humic substances during the decomposition of lignocellulose.

3.5. Electrochemical activity

The combination of intermittent micro-aeration and *P. chrysosporium* supplement enhanced the electrochemical activity of microbial communities. Cyclic voltammetry curves revealed that the sludge capacitance in the IMA-P group and IMA group reached 2.04×10^{-4} F and 1.84×10^{-4} F, respectively, which were 30.93 % and 17.70 % higher than the control group (1.56×10^{-4} F) (Fig. 6A). This suggested an increased availability of electron storage sites for microbial extracellular electron transfer [27]. Electrochemical impedance spectroscopy results further showed that the sludge resistance in the IMA-P and IMA groups were 1138 Ω and 2000 Ω, respectively, which were 53.93 % and 19.03 % lower than the control group (2470 Ω) (Fig. 6B). Lower resistance indicated the accelerated electron transfer within the sludge. Resistance and capacitance are commonly used indicators for evaluating extracellular electron transfer in microbial communities [39–41]. These results demonstrated that intermittent micro-aeration coupled with *P. chrysosporium* addition improved microbial electrochemical activity, likely attributed to the generation of electroactive substances (e.g., quinones) during the humification of organic substrates.

3.6. Microbial community

3.6.1. Proliferation of fungi

To verify whether the exogenous *P. chrysosporium* proliferated in the

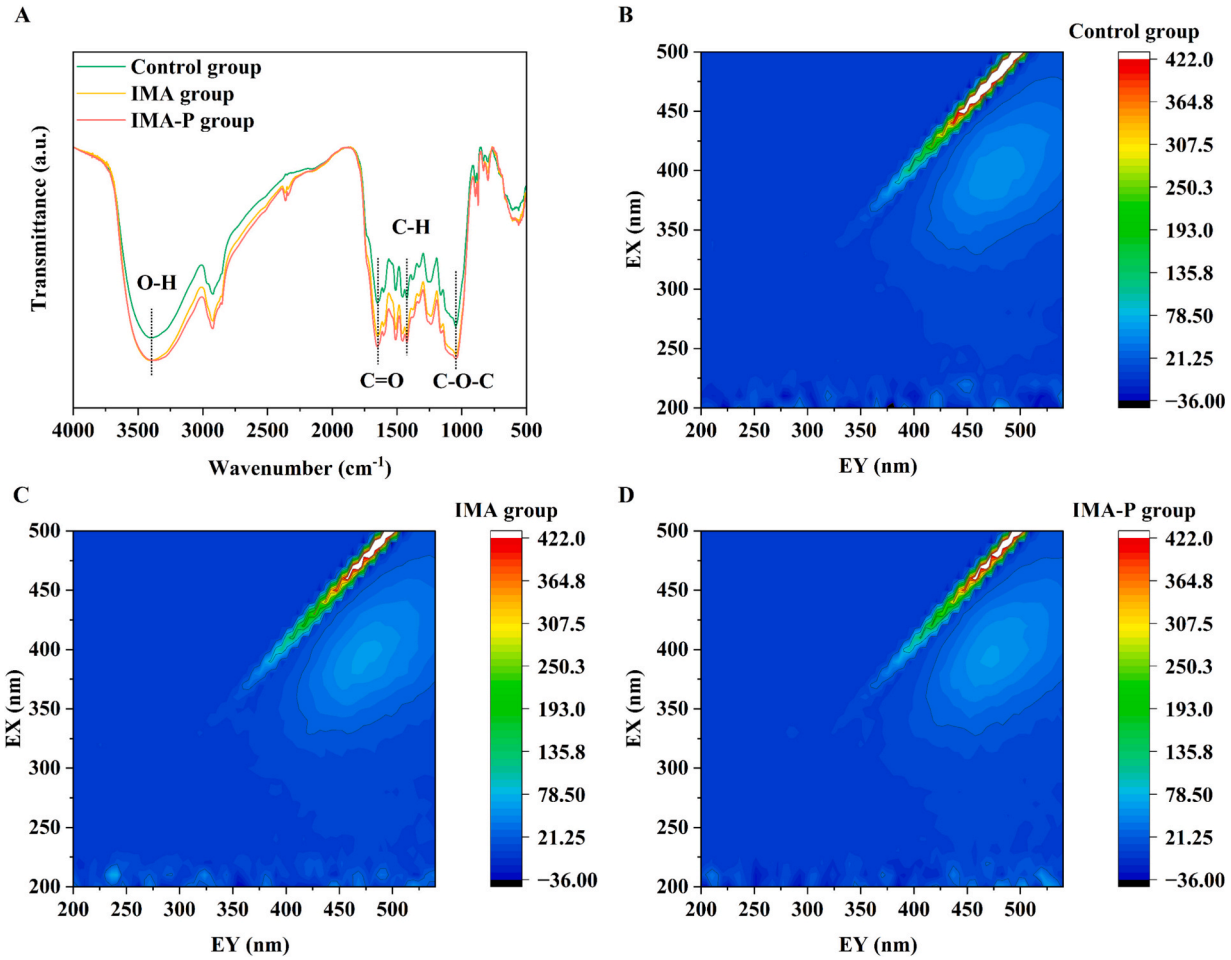


Fig. 5. Degradation of lignocellulose. (A) FTIR of sludge at the end of the experiment. (B, C, D) 3D-EEM of sludge samples from the control, IMA, and IMA-P group.

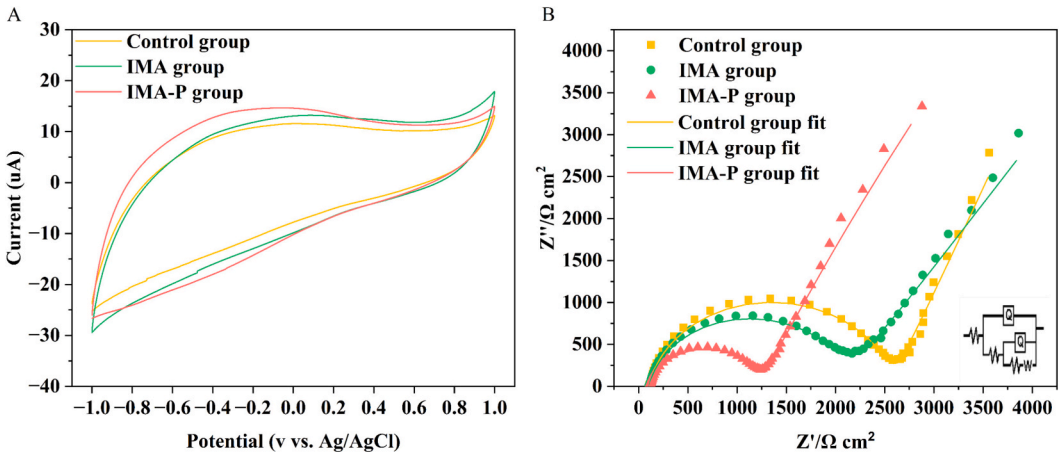


Fig. 6. Changes in microbial electrical activity. (A) Curve of cyclic voltammetry. (D) Curve of electrochemical impedance spectroscopy.

anaerobic system, the relative abundance of fungi in sludge was analyzed on days 36, 72, and 108. As shown in Fig. 7A, the relative abundance of *P. chrysosporium* in the fungal community reached 42.96 % at the end of the first stage (day 36). By the end of the second stage (day 72), its abundance decreased from an initial 9.55 % (theoretical value: 42.96 %/4.5, 4.5 representing the dilution by feeding) to 5.20 % under such intermittent micro-aeration (4 %($V_{Aeration}$:$V_{Reactor}$)/day). However, during the third stage (days 72–108), as the aeration intensity increased from 4 % to 8 % ($V_{Aeration}$:$V_{Reactor}$)/day, the relative abundance of *P. chrysosporium* rose from an initial 1.16 % (theoretical value: 5.20 %/4.5) to 1.54 %. In contrast, the relative abundance of *P. chrysosporium* in both the control group and IMA group was very low at the end of the operation on day 108 (<0.01 %). These results demonstrated that O_2 supply was critical for sustaining aerobic *P. chrysosporium* survival in anaerobic environments, and aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day could maintain its proliferation.

3.6.2. *Enrichment of dissimilatory iron-reducing bacteria*

The sludge collected from the reactors after 108 days of operation was analyzed through metagenomic analysis. Intermittent micro-aeration increased the diversity of the bacterial community (Table S3) and enriched dissimilatory iron-reducing bacteria (Fig. 7B), although

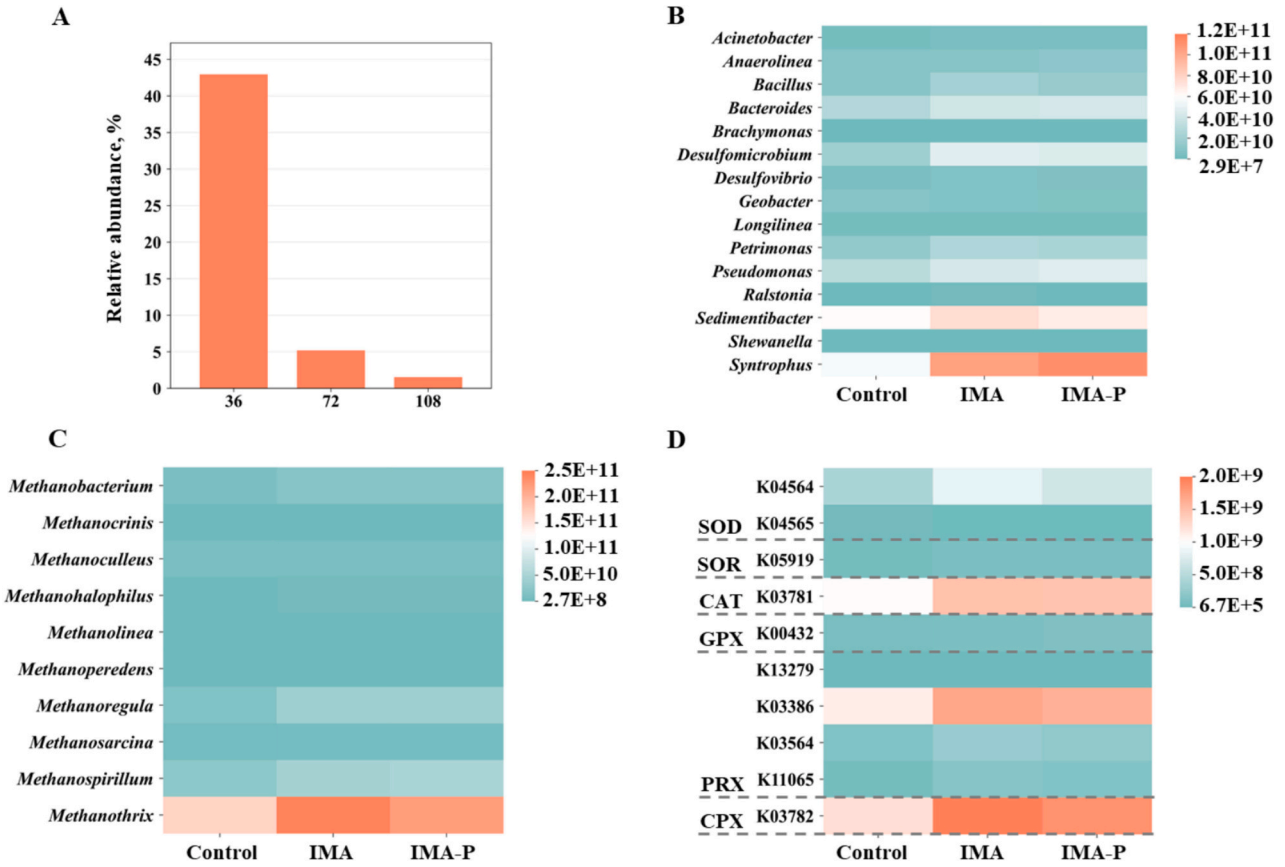


Fig. 7. (A) Relative abundance of fungi in the IMA-P group on days 36, 72, and 108. (B) Absolute abundance of bacteria on day 108. (C) Absolute abundance of archaeal on day 108. (D) Abundance of genes that encode the ROS detoxification in the methanogenic sludge. Superoxide oxide dismutase, SOD; superoxide reductase, SOR; catalase, CAT; glutathione peroxidase, GPX; peroxiredoxins, PRX; catalase-peroxidase, CPX.

there was no significant difference in the relative abundance of bacteria among the three groups (Fig. S2A). The total absolute abundance of dissimilatory iron-reducing bacteria (*Acinetobacter*, *Anaerolinea*, *Bacillus*, *Bacteroides*, *Brachymonas*, *Desulfomicrobium*, *Desulfovibrio*, *Geobacter*, *Longilinea*, *Petrimonas*, *Pseudomonas*, *Ralstonia*, *Sedimentibacter*, *Shewanella*, and *Syntrophus*) in the IMA-P group (3.81E+11 copies/g) and IMA group (3.82E+11 copies/g) was 1.61 and 1.62 times that in the control group (2.37E+11copies/g) [42,43]. Intermittent micro-aeration can oxidize Fe(II) into Fe(III), thus providing a continuous Fe(III) source for the dissimilatory iron reduction (DIR) process, and enabling the enrichment of dissimilatory iron-reducing bacteria [44,45]. Meanwhile, the abundances of DIR-related genes in the sludge of the IMA group and the IMA-P group were distinctly upregulated compared with those in the control group (Fig. S3), which indicated an enhanced extracellular respiration capacity in the groups with intermittent micro-aeration [46–48]. This result was consistent with the enhancement of electroactivity in the groups with intermittent micro-aeration.

3.6.3. Enrichment of methanogen

Intermittent micro-aeration increased the diversity of the bacterial community (Table S4). *Methanothrix* was identified as the predominant archaeal genus across all groups, demonstrating relative abundances of 26.91 % in the control group, 22.03 % in the IMA group, and 21.35 % in the IMA-P group, respectively, showing a declining trend (Fig. S2B). However, the absolute abundances of *Methanothrix* in IMA group (2.46E+11 copies/g) and IMA-P group (2.22E+11 copies/g) were much higher than those in the control group (1.67E+11 copies/g), being 1.47 times and 1.33 times that of the latter, respectively (Fig. 7C). Besides, as the methanogen genus with the second-highest relative abundance (4.06 %–4.79 %), *Methanospirillum* also had a higher absolute abundance in the IMA group (4.55E+10 copies/g) and IMA-P group (4.98E+10 copies/g), compared to the control group (2.52E+10 copies/g). This observed abundance of methanogens aligned with previous reports demonstrating an O_2 tolerance of both *Methanothrix* and *Methanospirillum*, suggesting their crucial role in maintaining methanogenic activity under micro-aeration conditions [49,50]. In addition, the ROS induced by intermittent micro-aeration accelerated the decomposition of lignocellulose, which could provide more available substrates for methanogens [17,45]. This might be the reason why the abundance of methanogens is higher in the groups with intermittent micro-aeration.

3.7. Antioxidant defense mechanisms

It has been commonly acknowledged that aerobic bacteria possess antioxidant defense systems, which are capable of efficiently counteracting and reducing the detrimental impacts stemming from an overaccumulation of ROS by synthesizing diverse antioxidant enzymes [51,52]. Conversely, obligate anaerobes do not have such a defense mechanism, making them unable to survive when exposed to oxidative stress [51,53]. Emerging research findings suggest that certain methanogenic genera can cope with oxidative stress by encoding antioxidant enzymes that serve to detoxify the cellular environment [54,55]. However, the reports on the antioxidant mechanisms of anaerobic microorganisms are still limited. In this study, the results of metagenomic analysis showed that the absolute abundance of superoxide oxide dismutase, superoxide reductase, catalase, glutathione peroxidase, peroxiredoxins, and catalase-peroxidase were obvious higher in both the IMA groups and IMA-P group compared to the control group (Fig. 7D). Specifically, the total abundance of antioxidant enzymes in the IMA groups (2.40E+09 copies/g) and the IMA-P group (2.19E+09 copies/g) were 1.73 and 1.57 times that of the control group (1.39E+09 copies/g), respectively. This result indicated that domesticated anaerobic sludge by intermittent micro-aeration could effectively scavenge intracellular $\bullet O_2^-$ and H_2O_2, protecting cells from oxidative stress damage [54].

The enrichment of methanogenic genera with the capacity to cope with O_2 stress, along with the elevated abundance of antioxidant enzyme genes, might account for the effective continuation of methanogenesis under intermittent micro-aeration conditions. This finding validated that the ability of anaerobic sludge to resist oxidative stress could be enhanced by O_2 induction within the anaerobic digestion system, which aligns with previous researches [45,54].

4. Conclusion

This study demonstrated that the *P. chrysosporium* supplementation coupled with intermittent micro-aeration induced ROS generation to accelerate organic solid waste decomposition and enhance methane production during anaerobic digestion. At an aeration of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day, the IMA and IMA-P groups achieved 17.22 % and 26.36 % higher methane production compared to the control, alongside the improvement of TS and VS removal. Intermittent micro-aeration triggered periodic redox oscillations to generate ROS. Under oxygen-limited conditions, *P. chrysosporium* secreted lignocellulolytic enzymes and enhanced •OH generation to accelerate lignocellulose degradation. Microbial community analysis confirmed *P. chrysosporium* proliferation in the intermittent micro-aerobic anaerobic system at an aeration intensity of 8 % ($V_{Aeration}$:$V_{Reactor}$)/day. This study elucidated the ROS-mediated enhancement of anaerobic digestion of organic solid waste by coupling *P. chrysosporium* supplementation with intermittent micro-aeration, providing novel insights for advancing micro-aerobic anaerobic digestion technology in engineering applications.

CRediT authorship contribution statement

Xuepeng Wang: Writing – original draft, Investigation, Formal analysis. **Zhenru Wang:** Investigation. **Ying Deng:** Methodology. **Haohao Mao:** Methodology. **Qilin Yu:** Writing – review & editing. **Yaobin Zhang:** Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.166351.

Data availability

Data will be made available on request.

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